5.4.1 Somatic Embryogenesis

Somatic embryogenesis is the formation of embryos from the callus tissue directly and these embryos are called **Embryoids** or from the *in vitro* cells directly form pre-embryonic cells which differentiate into embryoids.

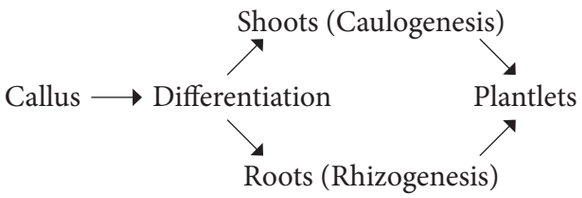
Applications

- Somatic embryogenesis provides potential plantlets which after hardening period can establish into plants.
- Somatic embryoids can be be used for the production of synthetic seeds.
- Somatic embryogenesis is now reported in many plants such as *Allium sativum*, *Hordeum vulgare, Oryza sativa, Zea mays* and this possible in any plant.

Synthetic seeds are produced by encapsulation of embryoids in agarose gel or calcium alginate.

5.4.2 Organogenesis

The morphological changes occur in the callus leading to the formation of shoot and roots is called organogenesis.



- Organogenesis can be induced *in vitro* by introducing plant growth regulators in the MS medium.
- Auxin and cytokinins induce shoot and root formation.

5.5 Applications of Plant Tissue Culture

Plant tissue culture techniques have several applications such as:

i. Improved hybrids production through somatic hybridization.
ii. Somatic embryoids can be encapsulated into synthetic seeds (synseeds). These encapsulated seeds or synthetic seeds help in conservation of plant biodiversity.
iii. Production of disease resistant plants through meristem and shoot tip culture.
iv. Production of stress resistant plants like herbicide tolerant, heat tolerant plants.
v. Micropropagation technique to obtain large numbers of plantlets of both crop and tree species useful in forestry within a short span of time and all through the year.
vi. Production of secondary metabolites from cell culture utilized in pharmaceutical, cosmetic and food industries.

Somaclonal variations: Somatic variations found in plants regenerated in vitro (i.e. variations found in leaf, stem, root, tuber or propagule)

Gametoclonal variations: Gametophytic variations found in plants regenerated in vitro gametic origin (i.e. variations found in gametes and gametophytes)

5.5.1 Micropropagation of Banana

Micropropagation of plants at industrial level maintains high standards of homogeneity in plants like pineapple, banana, strawberry and potato.



Figure 5.11: Micropropagation of Banana

5.5.2 Artificial Seed

Artificial seeds or synthetic seeds (synseeds) are produced by using embryoids (somatic embryos) obtained through in vitro culture. They may even be derived from single cells from any part of the plant that later divide to form cell mass containing dense cytoplasm, large nuclceus, starch grains, proteins, and oils etc., To prepare the artificial seeds different

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